

Principles of Economics

Problem Set 13

Universidad Carlos III de Madrid | Department of Economics

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1. WARMING-UP QUESTIONS

1. As we saw in Unit 3, technological progress increases your hourly productivity. This means that by working the same number of hours you could thus produce and consume more, or you can produce and consume the same amount of goods while working fewer hours and enjoying more free time.

The economist Oliver Blanchard argues that the difference in output per capita between the US and France is partially due to the fact that relative to those in the US, the French have used some of the increase in productivity to enjoy more free time rather than raise consumption.

Think about two countries, one that has lower GDP per capita due to fewer hours worked, and another that has higher GDP per capita due to more hours worked (such as France and the US). Assuming that overall life satisfaction consists only of free time and consumption, in which country would you expect overall life satisfaction to be higher, and why? Clearly state any assumptions you make about the preferences of residents in each country.

2. Considering only working hours and GDP per capita, which country (France or the US) would you prefer to live in, and why? How would your answer change if you considered other factors?
3. The next graph shows the unemployment rates for 16 OECD countries from 1960 to 2019.

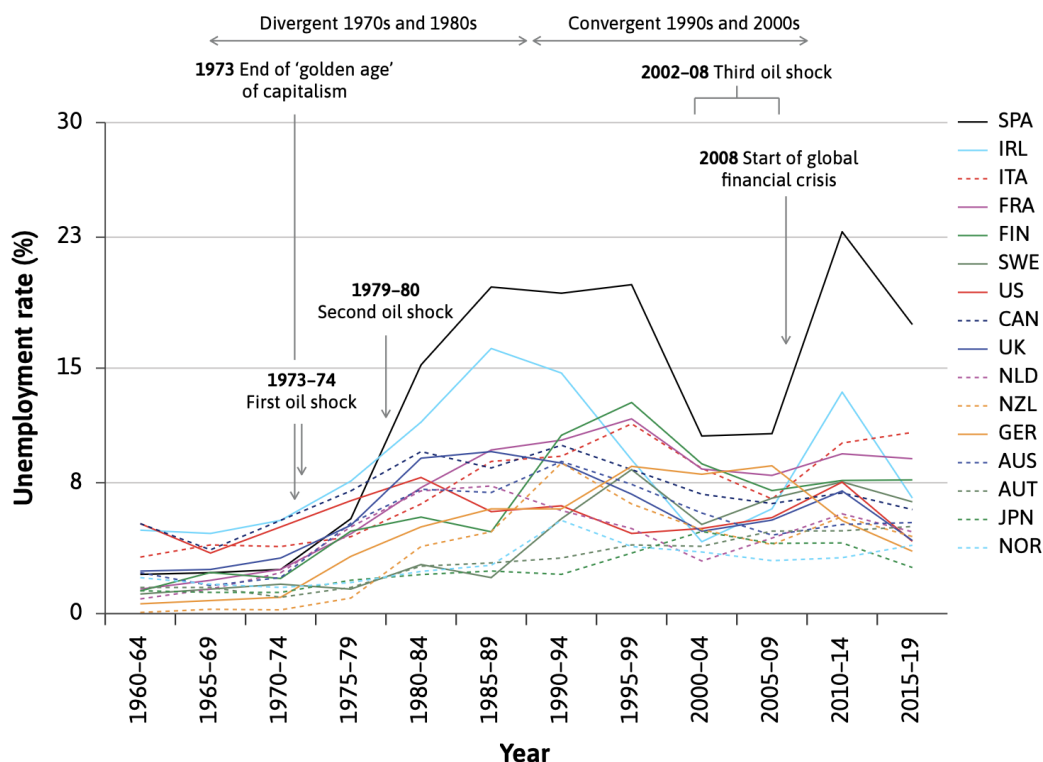
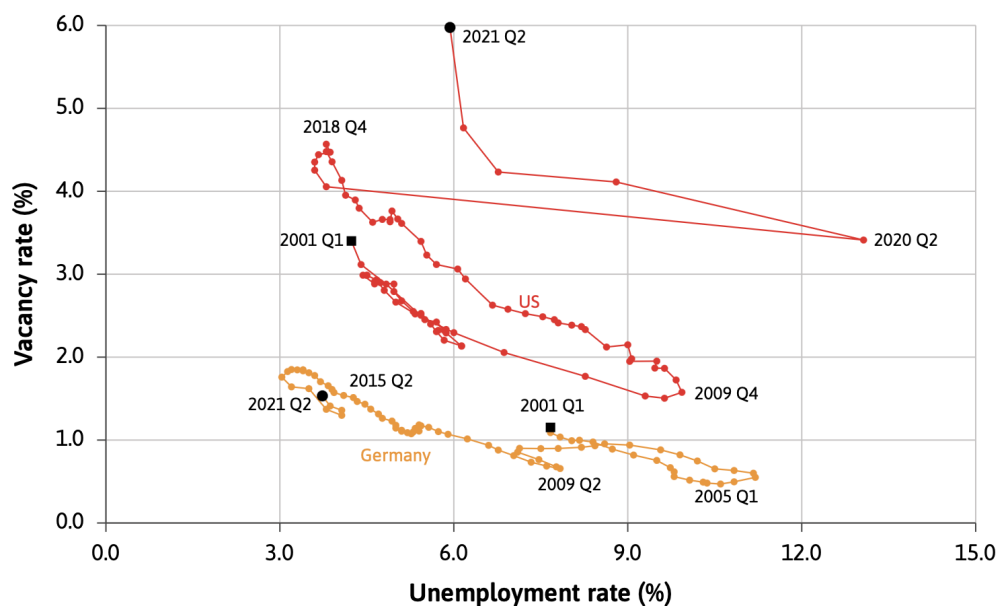


Figure 16.1 Unemployment rates in selected OECD countries (1960–2019).

Based on this information, which of the following statements is correct?

- There is no correlation between unemployment rates across countries.
 - There has been a clear upward trend in unemployment in all countries in the last 30 years.
 - The unemployment rates of different countries were affected very differently by the oil shocks of the 1970s.
 - The unemployment rate rose in all countries following the 2008 global financial crisis.
4. The graph shows the plot of Beveridge curves for the US and Germany for the period 2001 Q1 to 2021 Q2. Based on this information, which of the statements below is correct?



- The Beveridge curves depict the negative relationship between the vacancy rate and the employment rate.
- The US labour market was better at matching workers with vacancies during the financial crisis of 2008–9.
- The US Beveridge curve shifted after the financial crisis, improving the matching rate.
- The matching rate in Germany improved after its Beveridge curve shifted around 2007.

5. Refer to the next figure to answer the following questions:

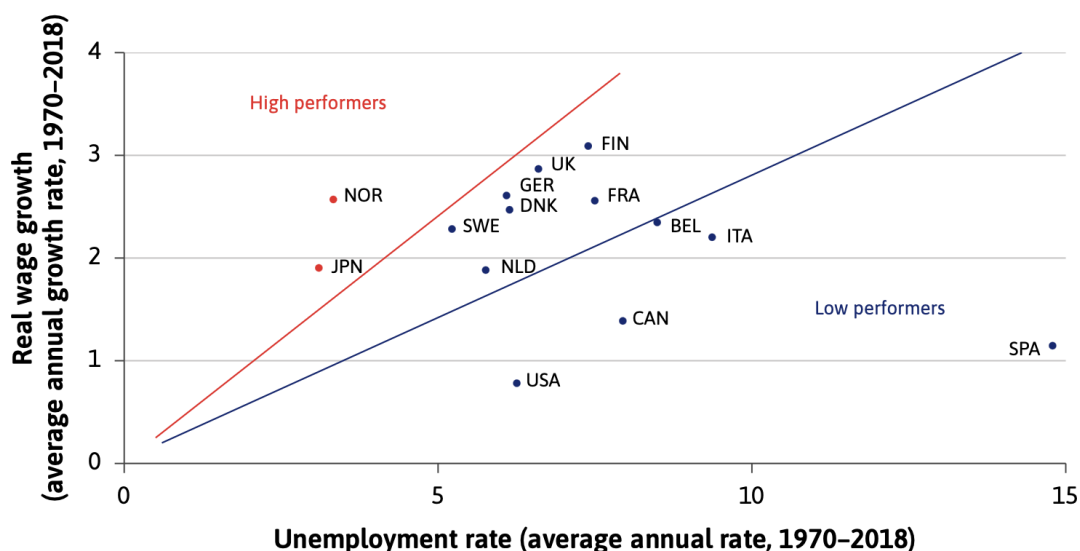


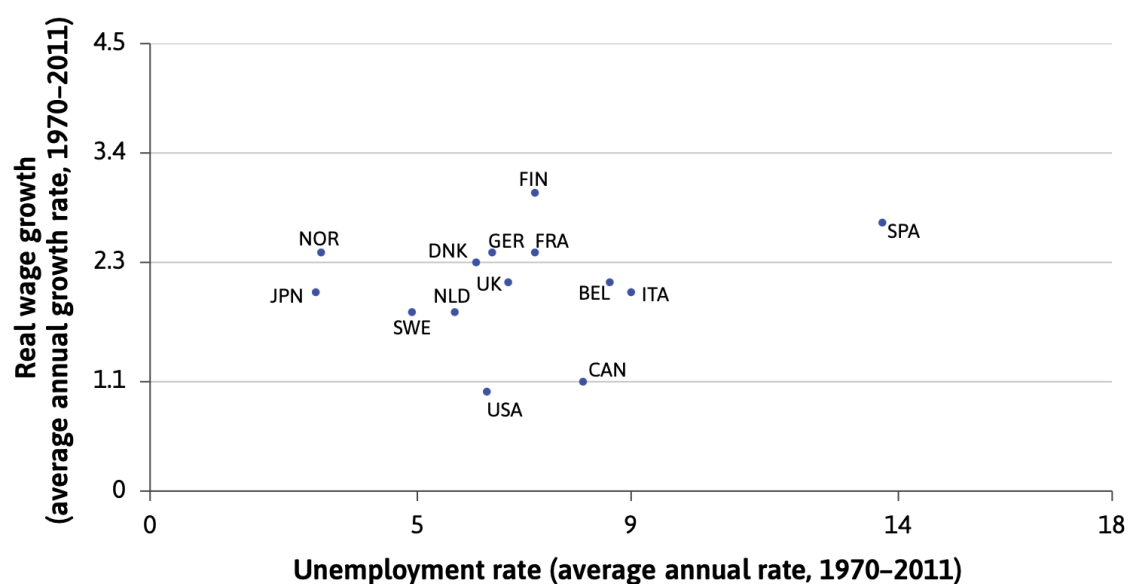
Figure 16.12 Long-run unemployment and real wage growth across the OECD (1970–2018).

- Using the same axes, draw the indifference curves of a citizen or policymaker who cares only about wage growth.
- According to the data in the figure, which countries would be the high performers and which would be the low performers?

- c. Using the same axes, draw your indifference curves if you only cared about the unemployment rate. Which countries would be the high and low performers in this case?
- d. Using the same axes, draw an indifference curve based on your own personal preferences over wage growth and unemployment, and justify your choice.
- e. Now considering your preferences over other economic factors, which country in the figure would you choose to live in, and why? Explain which economic factors you included in your decision.

2. MULTIPLE-CHOICE QUESTIONS

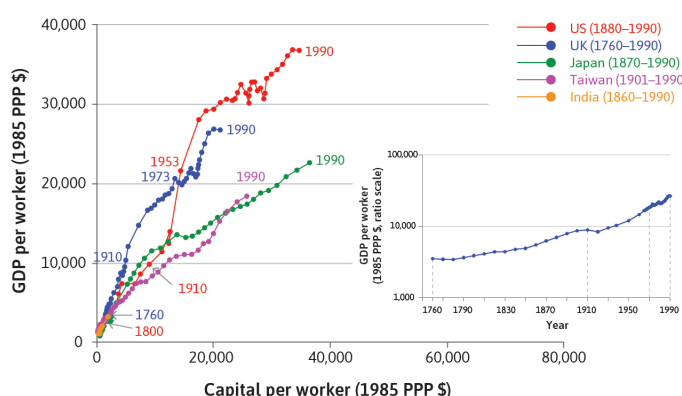
6. Which of the following statements is correct regarding the model of the labour market?
 - a) Labour-saving technological progress raises unemployment in both the short and long run.
 - b) In the long-run model, firms enter the market when the markup is low.
 - c) In the long-run model, the markup is independent of the number of firms.
 - d) In the short- and medium-run models the amount of capital is fixed, while in the long-run model the amount of capital can vary.
7. The following graph plots the real wage growth of different countries against their unemployment rate, averaged over the period 1970–2011.



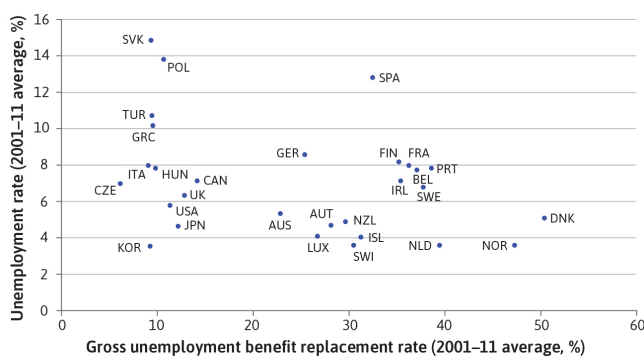
Based on this information, which of the following statements is correct?

- a) If you only cared about unemployment, then Finland is the country with the best performance.
- b) If you only cared about wage growth, then North American countries have outperformed European countries.
- c) If you cared only about wage growth, then Spain is one of the best-performing countries.
- d) If you cared about both unemployment and wage growth, then Finland has unambiguously outperformed Norway.

8. Consider a diagram with capital intensity on the horizontal axis and labour productivity on the vertical axis. Which of the following statements is/are true?
- Technological progress can be represented by an upward rotation of the production function.
 - If the production function is concave, the capital to output ratio declines as capital intensity rises.
 - In a stationary state as described by John Stuart Mill, growth in GDP per worker ceases because the production function shifts down.
9. The diagram plots GDP per worker vs. capital per worker, both across countries in 1990 (the right panel) and the trajectories since 1760 for a few representative countries (the left panel). Based on this information, which of the following statements is correct?

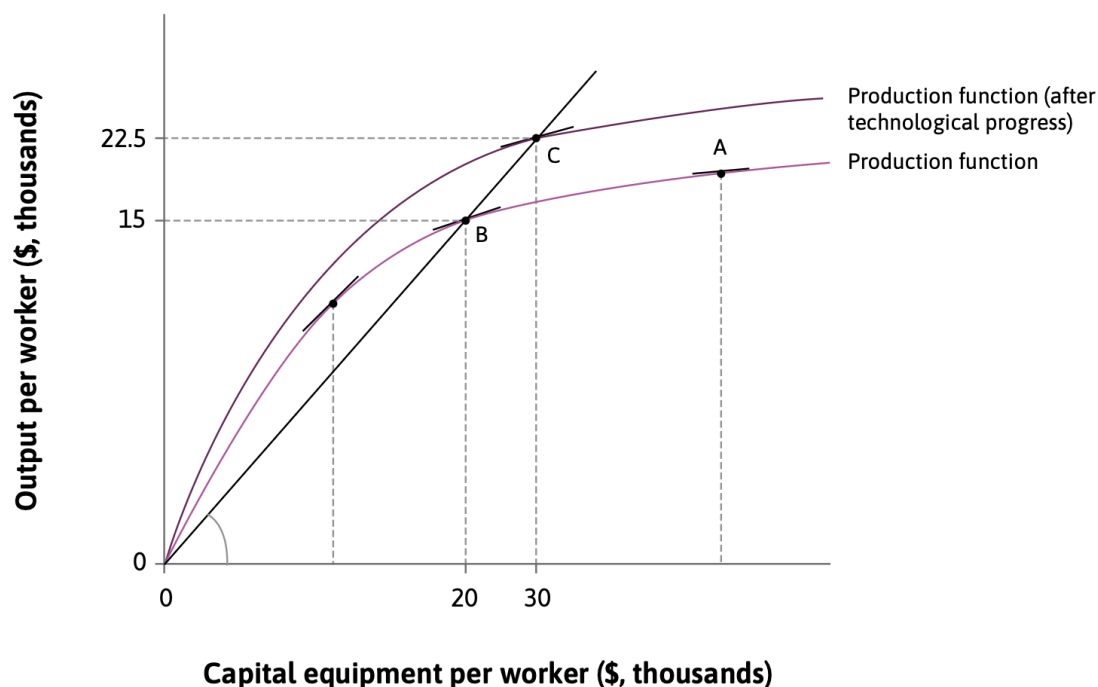


- There is no clear evidence of technological progress in GDP per worker in the US.
 - Taiwan was more capital intensive than the UK in 1990.
 - The average product of capital has been higher in Japan than in the UK over the years shown.
 - Japan has been the most successful country in attaining a high GDP per worker by use of its capital.
10. The figure shows a plot of unemployment rate vs. gross unemployment benefit replacement rate, averaged over the period 2001 to 2011. Based on this information, which of the following statements is correct?
- Eastern Asian countries achieved low unemployment rates despite having high unemployment benefits.
 - Data from Nordic countries suggests that high unemployment benefits lead to high unemployment rates.
 - The graph confirms the prediction of the labour market model that higher unemployment benefits would raise the wage-setting curve, leading to higher unemployment.
 - Countries that offer generous unemployment insurance schemes can still achieve low rates of unemployment.



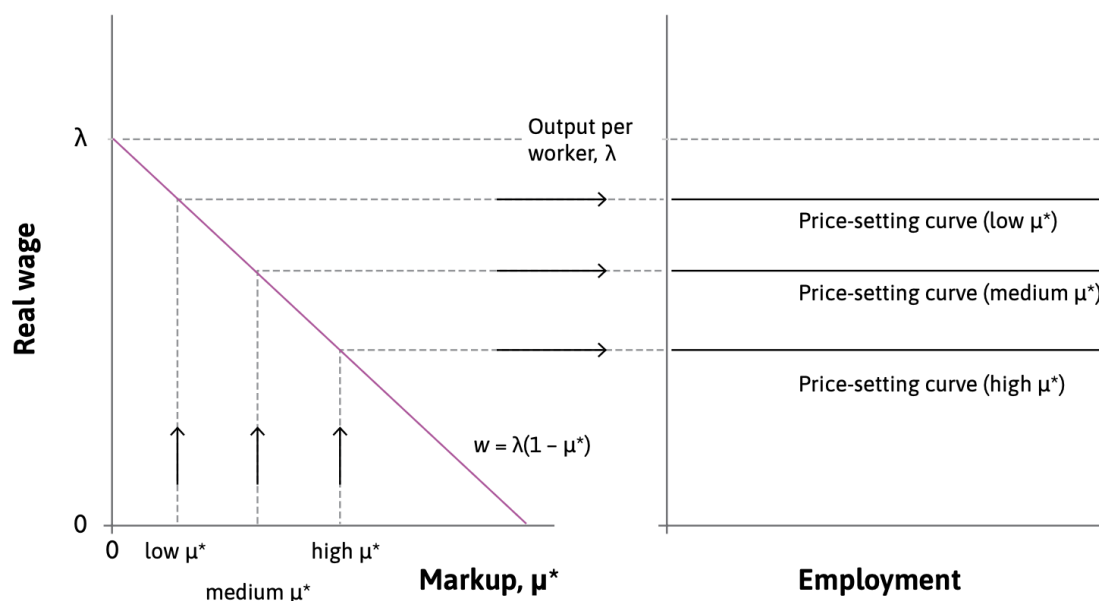
3. NUMERICAL PROBLEMS

11. The following diagram shows an economy's production function before and after technological progress:



Based on this information, which of the following statements is correct?

- a) The marginal product of capital at C is $(22,500 - 15,000) / (30,000 - 20,000) = 0.75$.
 - b) The concavity of the production function indicates increasing marginal product of capital.
 - c) As a result of a technological progress, the marginal product of capital rises but the average product of capital remains constant, for a given level of capital per worker.
 - d) The average product of capital at B is $15,000 / 20,000 = 0.75$.
12. The following graph depicts of the long-run price-setting curve and the markup at which firm entry and exit are both zero.

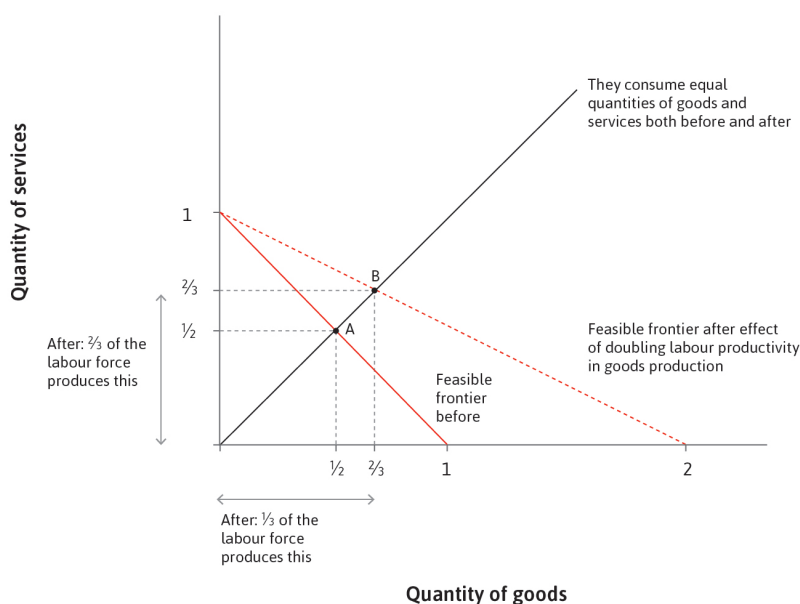


Based on this information, which of the following statements is correct?

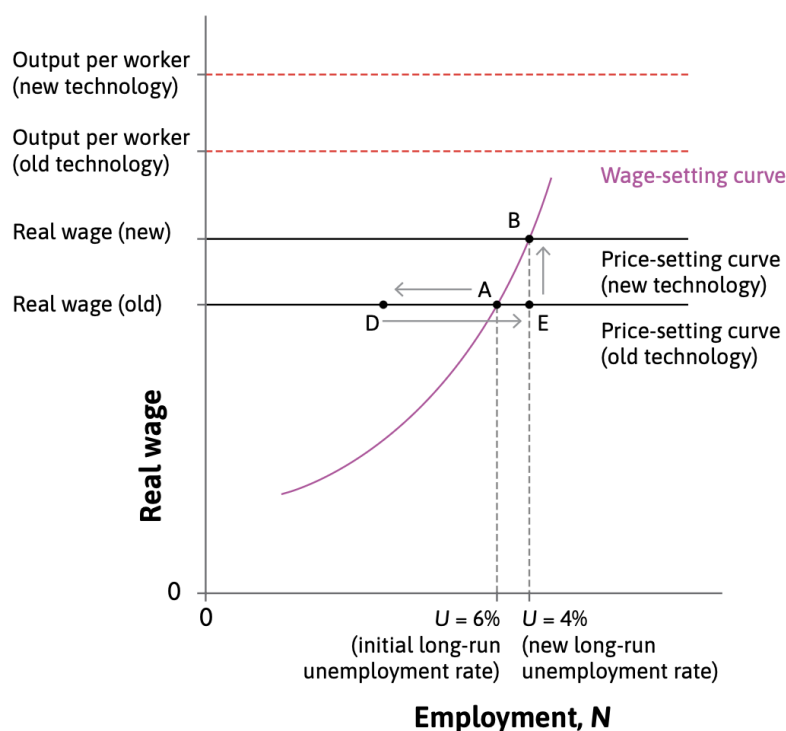
- a) An increase in the degree of competition in the economy will lower the price-setting curve.
- b) A lower interest rate leads to a lower price-setting curve.
- c) Lower worker productivity leads to a higher price-setting curve for a given markup μ^* .
- d) Higher risk of expropriation of businesses overseas results in a higher price-setting curve.

13. The figure is a model that explains how "deindustrialisation" of the workforce may occur due to an increase in the productivity of the goods sector. Before the change, the productivities in both sectors are 1 (i.e. if all labour is employed in one or the other sector then 1 unit of the output is produced). After the change, productivity in the goods sector doubles while productivity in the services sector remains constant. Finally, the population consumes an equal amount of goods and services at all times. Based on this information, which of the following statements is correct?

- a. After the productivity change, the population consumes 2/3 units each of goods and services. Therefore, the proportions employed in both sectors remain the same.
- b. The increase in the proportion of the workers in the services sector is caused by a fall in the absolute productivity in the sector.
- c. The assumption of faster productivity growth in the goods sector than in the services sector is unlikely in reality.
- d. In reality, the lower relative price of goods after the productivity rise is likely to lead to higher goods consumption, an effect which is not considered in the model above. This would somewhat offset the deindustrialisation effects demonstrated.



14. The figure below depicts the long-run adjustment process in the labour market after technological progress.



New technology shifts up output per worker and the price-setting curve

A→D: Introduction of a new technology leads to a rise in unemployment
D→E: High profits encourage new firms to enter
E→B: Lower unemployment leads to rising real wages
B: The new long-run rate of unemployment is 4%

Based on this information, which of the following statements is correct?

- The new technology does not cause any increase in unemployment, either in the short run or in the long run.
- At D firms increase investment, and hence employment, due to the large gap between the real wage paid and the workers' wage-setting curve.
- Lower unemployment at E implies a higher wage required to induce workers to exert high effort, resulting in the higher real wage at B.
- The adjustment from equilibrium A to the new equilibrium at B is immediate.

4. CONCEPTUAL QUESTIONS

15. What was the basis of US technological leadership:

- a) after the second world war
- b) from the mid-1990s (you may find sections 21.1 and 21.2 useful if you choose this option)?

Use the models you have learnt in the course (including readings suggested in the ebook and your own research) and in particular, make use of at least three of the following concepts, explaining their meaning, to justify your answer:

- production function; shift of and shift along
- labour-saving technology
- innovation rents
- research and development, education
- competition
- new entry.

Select one concept that is important to your explanation but is not in this list. You may refer to evidence and models presented in other units in The Economy. Be sure to provide appropriate referencing.

16. Based on the definition of creative destruction of Unit 2 by Joseph Schumpeter, give examples of destruction and creation, and identify the winners and the losers in the short and long run.

17. According to the Beveridge curves, the German labour market does a better job at matching workers with job openings, but over some intervals (for example, 2001 Q1 to 2005 Q1), average unemployment in Germany in Figure 16.6 was higher than in the US.

Consider the possible role of aggregate demand (Section 13.2 on Okun's law, and Section 14.10 on aggregate demand and unemployment). What kind of data could be used to find support for your hypothesis?